

■ Cloud Computing

Complete Question Bank — Answers

Section A (2 Marks) · Section B (5 Marks) · Section C (10 Marks)

SECTION A — Short Answer Questions (2 Marks each)

1. Differentiate between Grid Computing and Cluster Computing.

Feature	Grid Computing	Cluster Computing
Definition	Connects geographically distant computers	Connects nearby computers in one location
Location	Distributed (across internet)	Same building / LAN
Coupling	Loosely coupled	Tightly coupled
Goal	Share resources globally	High performance tasks
Example	SETI@home, LHC Grid	Beowulf cluster, Hadoop cluster

2. Why is Distributed Computing scalable?

Distributed computing is scalable because you can simply add more computers (nodes) to the network to increase capacity. No single machine needs to be upgraded — just add more machines as workload grows.

3. What is resource pooling?

Resource pooling means the cloud provider combines computing resources (CPU, memory, storage) and serves multiple customers from this shared pool. Resources are dynamically assigned based on demand.

4. State one limitation of Cluster Computing.

All nodes must be located at the same place (same LAN), making it difficult to scale across different geographic locations.

5. How does Grid Computing share resources?

Grid Computing shares resources over the internet using middleware software (like Globus Toolkit). Computers from different organizations join a virtual grid and contribute idle resources for shared tasks.

6. Mention two business drivers for Cloud adoption.

- Cost Reduction – Pay only for what you use; no need to buy expensive hardware.
- Scalability – Easily scale up or down based on business needs.

7. Define Cloud Computing (NIST – any two aspects).

NIST defines Cloud Computing as a model for enabling convenient, on-demand network access to a shared pool of configurable resources. Two key aspects:

- On-demand self-service – Users can get resources automatically without human interaction.
- Broad network access – Services available over the network via standard devices.

8. Name two cloud service providers.

- Amazon Web Services (AWS)
- Microsoft Azure

9. State two characteristics of Cloud Computing.

- Elasticity – Resources can be scaled up or down quickly.
 - Measured Service – Usage is monitored and billed automatically (pay-as-you-go).
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10. Mention one disadvantage of Cloud Computing.

Vendor Lock-in: Once you move your data and applications to a cloud provider, it becomes difficult and costly to switch to another provider.

11. Compare Cloud and Grid Computing (one point).

Cloud Computing	Grid Computing
Offers services (IaaS, PaaS, SaaS) on demand	Shares resources across organizations for scientific tasks
Managed by a single provider	Managed by multiple organizations collectively

12. Role of open standards in the cloud.

Open standards ensure interoperability between different cloud platforms, prevent vendor lock-in, and allow applications to work across multiple cloud providers using common APIs and protocols (e.g., OpenAPI, OVF).

13. Mention two benefits of cloud computing.

- Cost Savings – No need to invest in physical hardware; pay only for usage.
 - Anywhere Access – Access data and applications from any device, anywhere.
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14. Define Cloud Computing Stack.

The Cloud Computing Stack refers to the three layers of cloud services stacked on top of each other: IaaS (Infrastructure) at the bottom, PaaS (Platform) in the middle, and SaaS (Software) at the top.

15. What is a Hypervisor?

A Hypervisor (also called Virtual Machine Monitor) is software that creates and manages Virtual Machines (VMs) on a physical server. It allows multiple VMs to run on one physical machine. Types: Type 1 (bare-metal, e.g., VMware ESXi) and Type 2 (hosted, e.g., VirtualBox).

16. Role of networks in the cloud.

Networks are the backbone of cloud computing. They connect users to cloud servers, enable data transfer between cloud resources, and provide services like load balancing, VPNs, and CDNs to ensure fast and secure access.

17. Name two cloud protocols.

- SOAP (Simple Object Access Protocol) – XML-based protocol for web services.
 - REST (Representational State Transfer) – Lightweight HTTP-based protocol used in modern APIs.
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18. Function of web services.

Web services allow different applications to communicate with each other over the internet using standard protocols (HTTP, XML, JSON). In cloud, they expose APIs so developers can use cloud features programmatically.

19. IaaS vs PaaS (two differences).

Feature	IaaS	PaaS
What it provides	Raw infrastructure (VMs, storage, networking)	Platform with tools for app development
User manages	OS, middleware, runtime, apps	Only the application and data
Example	AWS EC2, Azure VMs	Google App Engine, Heroku

20. Public vs Private Cloud (two differences).

Feature	Public Cloud	Private Cloud
Ownership	Owned by cloud provider	Owned by single organization
Security	Shared, less control	Dedicated, high control
Cost	Low (pay-as-you-go)	High (own hardware)

21. What is cloud billing?

Cloud billing is the system that tracks your cloud resource usage and charges you accordingly. It uses the pay-as-you-go model — you pay only for what you use (per hour, per GB, per request).

22. What is vendor lock-in?

Vendor lock-in occurs when a customer becomes so dependent on a cloud provider's proprietary tools and services that switching to another provider becomes very difficult, costly, or technically complex.

23. Name any two cloud service providers.

1) Amazon Web Services (AWS) 2) Google Cloud Platform (GCP)

24. What is scalability in cloud computing?

Scalability is the ability to increase or decrease computing resources (CPU, memory, storage) easily based on demand — ensuring performance is maintained as workload changes.

25. Differentiate between vertical scaling and horizontal scaling.

Feature	Vertical Scaling (Scale Up)	Horizontal Scaling (Scale Out)
Meaning	Add more power to existing machine	Add more machines to the pool
Example	Upgrade 4GB RAM to 16GB RAM	Add 5 more servers
Limit	Hardware has a physical limit	No practical limit
Downtime	May require restart	Usually no downtime

26. What is pay-as-you-go model?

Pay-as-you-go means you pay only for the cloud resources you actually use. Like a utility bill — if you use 10 hours of a server, you pay for 10 hours only. No upfront cost, no waste.

27. Define data privacy in cloud computing.

Data privacy in cloud refers to protecting personal and sensitive data stored in cloud from unauthorized access, ensuring it is used only for its intended purpose and complies with privacy laws (like GDPR).

28. What are jurisdictional issues in data storage?

When data is stored in a cloud server in another country, the laws of that country apply to the data. This creates jurisdictional issues — different countries have different privacy and security laws, making compliance difficult.

29. What is Identity and Access Management (IAM)?

IAM is a framework that controls who can access cloud resources and what they can do. It manages user identities, authentication (verifying who you are), and authorization (what you are allowed to do).

30. Define authentication in cloud computing.

Authentication is the process of verifying a user's identity before giving access to cloud resources. Methods include passwords, OTP, biometrics, and multi-factor authentication (MFA).

31. What is access control?

Access control defines what resources a verified user can access and what operations (read, write, delete) they can perform. It enforces the principle of least privilege — give only the minimum access needed.

32. What is data storage security?

Data storage security involves protecting data stored in cloud from unauthorized access, corruption, or loss. Techniques include encryption at rest, access controls, data replication, and regular backups.

33. What is Eucalyptus?

Eucalyptus (Elastic Utility Computing Architecture for Linking Your Programs To Useful Systems) is an open-source cloud computing platform that allows organizations to build private clouds compatible with AWS APIs.

34. What are the components of a Virtual Machine?

A Virtual Machine (VM) consists of: Virtual CPU (vCPU), Virtual Memory (RAM), Virtual Storage (virtual hard disk), Virtual Network Interface Card (vNIC), and a Guest OS running inside it.

35. What is Storage in PaaS?

PaaS provides storage services like databases (SQL/NoSQL), object storage, and file storage as managed services. Developers use these without managing the underlying hardware or software — e.g., Google Cloud SQL, Firebase.

SECTION B — Medium Answer Questions (5 Marks each)

1. Explain Grid, Cluster, and Distributed Computing with examples.

Cluster Computing: Multiple computers (nodes) connected in the same location via LAN to work together as one powerful system. Example: University computing lab running parallel simulations.

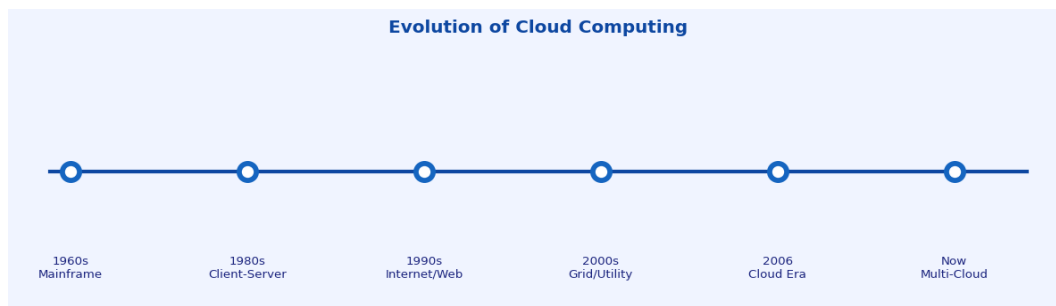
Grid Computing: Connects computers across different locations/organizations over the internet to share resources for large tasks. Example: SETI@home (searching for alien signals using volunteer computers worldwide).

Distributed Computing: A broader concept where multiple independent computers communicate and coordinate to complete a task. Example: Google's search engine uses thousands of distributed servers.

2. Discuss the evolution of Cloud Computing.

Cloud computing evolved through several stages:

- 1960s – Mainframe Computing: Large central computers shared by multiple users via terminals.
- 1980s – Client-Server Model: Personal computers connected to servers in local networks.
- 1990s – Internet Era: Web technologies enabled remote access to services.
- Early 2000s – Grid & Utility Computing: Resources shared like electricity (pay per use).
- 2006 – Cloud Era: Amazon launched AWS EC2, the first commercial cloud service.
- 2010s–Now – Multi-Cloud & Hybrid: Organizations use multiple providers; AI/ML on cloud.



3. Explain Utility Computing and compare it with Cloud Computing.

Utility Computing: A model where computing resources (CPU, storage, bandwidth) are provided as a metered service — like electricity or water. You pay only for what you use.

Feature	Utility Computing	Cloud Computing
Focus	Metered billing model	Full service delivery model
Services	Basic compute resources	IaaS + PaaS + SaaS
Self-service	Limited	Full on-demand self-service
Access	Often internal/VPN	Over public internet
Example	Grid usage billing	AWS, Azure, GCP

4. Describe business drivers for cloud adoption.

- **Cost Reduction:** Eliminates need to buy and maintain expensive hardware.
- **Scalability:** Scale resources up/down as per business demand.

- **Agility:** Launch new services quickly without infrastructure delays.
- **Disaster Recovery:** Cloud provides easy backup and recovery solutions.
- **Global Reach:** Deploy applications worldwide instantly.

5. Define Cloud Computing (NIST) and explain characteristics.

NIST Definition: Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned with minimal management effort.

Five Essential Characteristics:

- **On-demand Self-service:** Users provision resources automatically.
- **Broad Network Access:** Available via internet on any device.
- **Resource Pooling:** Resources shared among multiple customers.
- **Rapid Elasticity:** Resources scale up/down quickly.
- **Measured Service:** Usage tracked and billed automatically.

6. Discuss advantages and disadvantages of cloud computing.

Advantages	Disadvantages
Cost savings – no hardware purchase	Vendor lock-in – hard to switch providers
Scalability – grow/shrink instantly	Internet dependency – needs constant connectivity
Anywhere access – any device	Security concerns – data stored externally
Automatic updates – provider handles it	Limited control – provider manages infrastructure
Disaster recovery – built-in backup	Downtime – provider outages affect users

7. Compare Cloud, Grid, and Cluster Computing.

Feature	Cloud	Grid	Cluster
Location	Internet (remote)	Multi-org, distributed	Single location (LAN)
Ownership	Cloud provider	Multiple organizations	Single organization
Scalability	Very high	High	Limited by hardware
Service model	IaaS/PaaS/SaaS	Resource sharing	High-performance tasks
Example	AWS, Azure	SETI@home, LHC	Beowulf cluster

8. Explain the role of open standards.

Open standards are publicly available specifications that ensure different cloud systems can work together. Their role:

- **Interoperability:** Allows apps to work across multiple cloud platforms.
- **Avoid Lock-in:** Easy to migrate between providers.
- **Security:** Standard encryption protocols (TLS, OAuth) protect data.
- **Cost Reduction:** Open-source tools reduce licensing costs.
- **Examples:** OpenAPI, OVF (Open Virtualization Format), CDMI.

9. Explain Cloud Computing Architecture.

Cloud computing architecture consists of two main parts:

Front End: Client-side — includes the user's device (PC, mobile) and the browser or application used to access cloud services.

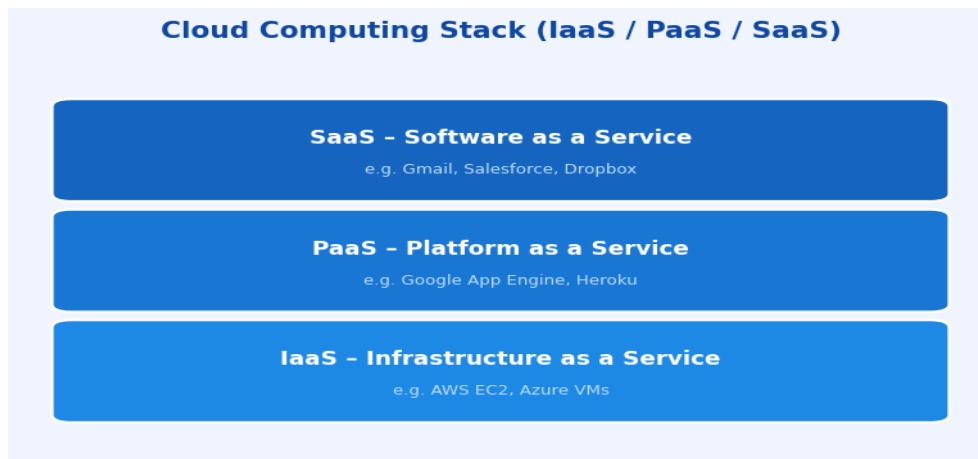
Back End: Server-side — includes cloud servers, storage, databases, security mechanisms, and a central server that manages communication.

Both are connected via the Internet. The back end has: Servers, Virtualization layer (Hypervisor), Storage, Security, Management software.

10. Describe Cloud Computing Stack.

The cloud stack has 3 layers — each builds on the one below it:

- **IaaS (Infrastructure as a Service):** Provides raw infrastructure — VMs, storage, networking. User manages OS and above. Example: AWS EC2.
- **PaaS (Platform as a Service):** Provides development platform with tools, databases, runtime. User manages only the application. Example: Google App Engine.
- **SaaS (Software as a Service):** Provides ready-to-use software. User just uses it. Example: Gmail, Salesforce.



11. Explain how cloud computing works.

Cloud computing works in these steps:

- User sends a request via browser/app over the internet.
- Request reaches the cloud provider's data center through the network.
- The load balancer distributes the request to the appropriate server.
- Virtualization software creates/manages virtual machines to process the request.
- Result is sent back to the user over the internet.

Key components: Internet, Load Balancer, Hypervisor, VMs, Storage, Databases.

12. Limitations of IaaS, PaaS, SaaS.

IaaS Limitations	PaaS Limitations	SaaS Limitations
User must manage OS and middleware	Limited control over infrastructure	Little to no customization
Requires technical expertise	Vendor lock-in due to specific tools	Data stored with provider — privacy concern

IaaS Limitations	PaaS Limitations	SaaS Limitations
Security depends on user config	Performance depends on provider	Dependent on internet connectivity

13. Describe how billing and accounting work in cloud computing.

Cloud billing works on the **pay-as-you-go** model:

- **Metering:** Every resource (CPU hours, storage GB, data transfer) is measured.
- **Pricing:** Provider charges per unit (per hour, per GB, per API call).
- **Billing Cycle:** Monthly invoices generated automatically.
- **Budget Alerts:** Users can set spending limits and alerts.
- **Tools:** AWS Cost Explorer, Azure Cost Management, GCP Billing.

14. Compare traditional hardware scaling with cloud-based scaling.

Aspect	Traditional Scaling	Cloud Scaling
Speed	Weeks (procure hardware)	Minutes (click a button)
Cost	High upfront capital	Pay only when needed
Direction	Only scale up (more HW)	Up and down, in and out
Waste	Over-provisioned capacity	Precise — scale to demand
Downtime	Often required	Typically zero-downtime

15. Explain different types of cloud databases and data storage services.

- **Relational DB (SQL):** Structured tables with SQL queries. Example: Amazon RDS, Cloud SQL.
- **NoSQL DB:** Unstructured/semi-structured data. Example: MongoDB on Atlas, DynamoDB.
- **Object Storage:** Store large unstructured data (images, videos). Example: AWS S3.
- **Block Storage:** Like a hard drive attached to a VM. Example: AWS EBS.
- **File Storage:** Shared file systems. Example: Amazon EFS, Azure Files.

16. Write a short note on trust, reputation, and risk in cloud computing.

Trust: Users must trust cloud providers to keep data secure, available, and private. SLAs (Service Level Agreements) help formalize this trust.

Reputation: A provider's track record (uptime, security incidents, compliance) determines its reputation. Providers like AWS and Azure have built strong reputations.

Risk: Cloud risks include data breaches, outages, compliance violations, and vendor lock-in. Risk mitigation strategies: encryption, multi-cloud, regular audits, backup plans.

17. Explain PaaS architecture.

PaaS sits between IaaS and SaaS. Its architecture has these layers:

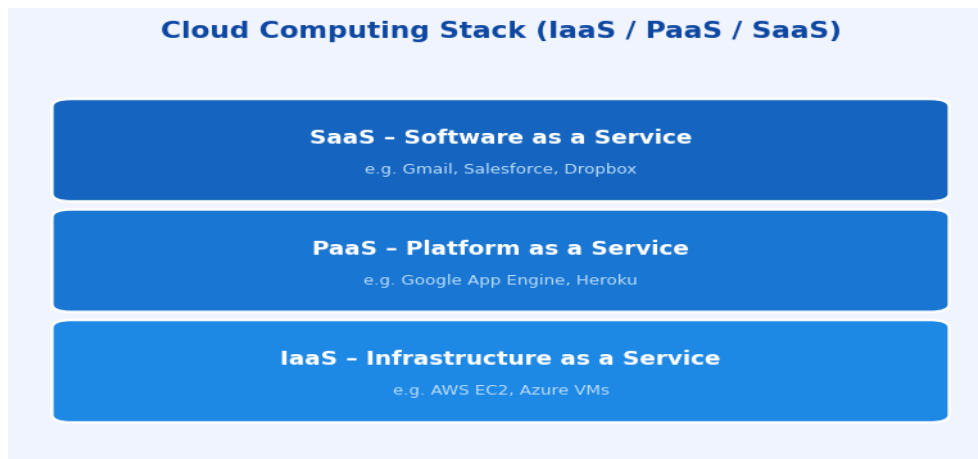
- **Infrastructure Layer:** Underlying VMs, servers (managed by provider).
- **Middleware Layer:** OS, web server, runtime environment (managed by provider).
- **Application Layer:** Developer writes code and deploys it here.
- **Database Services:** Managed database services provided.
- **User Layer:** End users access the deployed application.

Examples: Heroku (push code → it deploys automatically), Google App Engine.

18. Explain the cloud computing stack in detail with neat diagram.

The cloud stack layers, from bottom to top:

- **IaaS:** Provides virtualized computing infrastructure. Users deploy VMs, configure storage and networking. Full OS control. Best for: system admins, developers needing custom OS.
- **PaaS:** Provider manages everything up to runtime. Developer just writes and deploys code. Best for: application developers who want to focus on code, not infrastructure.
- **SaaS:** Fully managed application delivered over internet. No installation. Best for: business users who just need to use the software.



19. Explain the concept of virtualization in cloud computing.

Virtualization is the core technology of cloud computing. It creates virtual versions of physical resources:

- **Server Virtualization:** One physical server runs multiple virtual machines (VMs).
- **Storage Virtualization:** Multiple storage devices appear as one logical unit.
- **Network Virtualization:** Virtual networks (VLANs, VPNs) created on physical network.

How it works: A Hypervisor software (VMware, Hyper-V, KVM) is installed on the physical server. It creates isolated VMs, each with its own OS, resources, and applications. This enables resource pooling, isolation, and efficient utilization.

20. Compare open-source cloud platforms like Eucalyptus with commercial platforms like Amazon EC2.

Feature	Eucalyptus (Open-source)	Amazon EC2 (Commercial)
Cost	Free (open-source)	Pay-as-you-go pricing
Control	Full control on your hardware	Limited control; AWS manages infra
Compatibility	AWS-compatible APIs	AWS native
Scalability	Limited by your hardware	Virtually unlimited
Support	Community support	24/7 enterprise support
Use case	Private cloud / research	Production enterprise cloud

SECTION C — Long Answer Questions (10 Marks each)

1. Explain Grid Computing, Cluster Computing, and Distributed Computing in detail. Compare their architecture, advantages, and limitations with suitable examples.

1. Cluster Computing

A cluster is a group of computers (nodes) connected via high-speed LAN in the same physical location, working together as a single system.

- Architecture: Master node controls worker nodes. All use same OS and hardware (homogeneous).
- Advantages: High performance, fault tolerance, load balancing.
- Limitations: Expensive to set up, limited to one location, homogeneous requirement.
- Example: Beowulf cluster for weather simulation.

2. Grid Computing

A grid connects computers across multiple locations, organizations, and even countries using the internet, forming a virtual supercomputer.

- Architecture: Loosely coupled, heterogeneous, uses middleware like Globus Toolkit.
- Advantages: Uses idle computing worldwide, very large scale, resource sharing.
- Limitations: High latency, complex coordination, security challenges.
- Example: CERN's LHC Computing Grid for particle physics.

3. Distributed Computing

A broad model where multiple independent computers communicate and coordinate to solve a problem. Both cluster and grid are types of distributed computing.

- Architecture: Peer-to-peer or client-server, message passing (MPI, RPC).
- Advantages: Fault tolerance, scalability, geographic distribution.
- Limitations: Synchronization complexity, partial failures, network dependency.
- Example: Google's MapReduce, Bitcoin network.

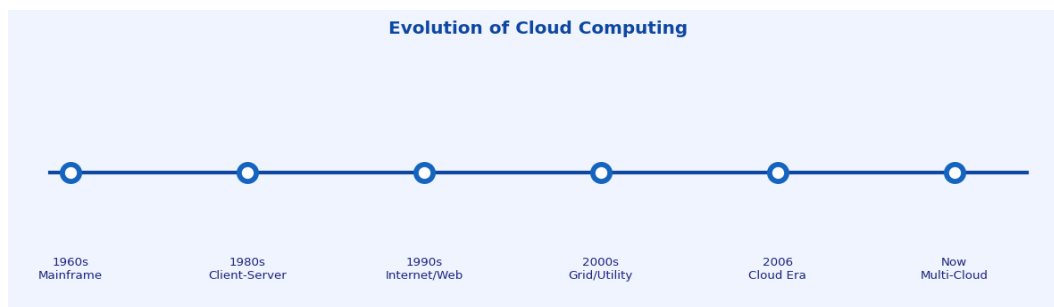
Feature	Cluster	Grid	Distributed
Location	Same site	Multi-site	Any location
Coupling	Tight	Loose	Loose
OS	Homogeneous	Heterogeneous	Heterogeneous
Scale	Medium	Large	Very large
Example	Beowulf	LHC Grid	Google, Bitcoin

2. Discuss the evolution of Cloud Computing from traditional computing paradigms. Explain the key business drivers that led to its rapid adoption.

Evolution Phases:

- **Mainframe Era (1960s):** Centralized computing, dumb terminals, expensive.
- **Client-Server Era (1980s):** Personal computers, local servers, distributed apps.
- **Internet Era (1990s):** Web, email, basic online services. ISPs emerged.

- **Grid/Utility Computing (2000s):** Resources treated like utilities (pay-per-use).
- **Cloud Era (2006):** Amazon launched AWS. Elastic, on-demand computing born.
- **Modern Era (2010s-now):** Serverless, containers, AI/ML services on cloud.



Key Business Drivers:

- **Cost Reduction:** No CapEx on hardware; only OpEx on usage.
- **Time to Market:** Launch services in hours instead of months.
- **Globalization:** Reach customers worldwide from day one.
- **Innovation:** Access to AI, ML, Big Data tools instantly.
- **Disaster Recovery:** Built-in backup across multiple data centers.
- **Remote Work:** Pandemic accelerated cloud adoption for distributed teams.

3. Define Cloud Computing based on the NIST model. Explain all essential characteristics and discuss their significance in real-world applications.

NIST Definition: Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Five Essential Characteristics:

Characteristic	Meaning	Real-world Significance
On-demand Self-service	Provision resources without human interaction	Startup launches server at midnight without calling AWS
Broad Network Access	Access via internet on any device	Employee works from home on any device
Resource Pooling	Resources shared among multiple customers	AWS serves millions from same data center
Rapid Elasticity	Scale up/down quickly	E-commerce site handles Black Friday traffic spike
Measured Service	Usage tracked and billed automatically	Company pays exactly for 1000 API calls, not more

4. Describe the Cloud Computing Architecture in detail. Compare it with traditional client-server architecture and explain how cloud improves performance and scalability.

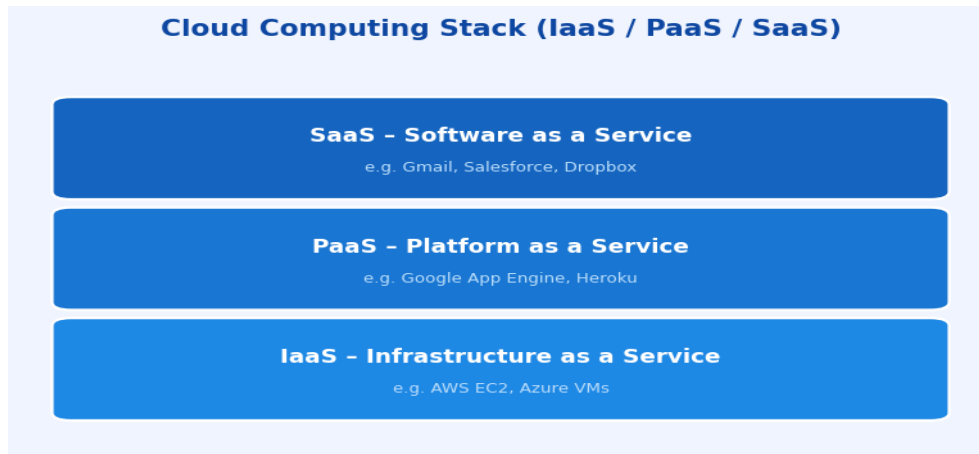
Cloud Computing Architecture:

- **Front End:** Client's device + browser/app. Sends requests over internet.
- **Internet:** The communication medium connecting front and back end.
- **Back End (Cloud):**

- **Load Balancer:** Distributes incoming requests across servers.
- **Servers:** Physical machines running hypervisors.
- **Hypervisor:** Creates and manages virtual machines.
- **VMs:** Isolated instances for different users/apps.
- **Storage:** Distributed storage systems (object, block, file).
- **Database:** Managed SQL/NoSQL services.
- **Security:** Firewalls, IAM, encryption.

Feature	Traditional Client-Server	Cloud Architecture
Servers	Fixed number, on-premise	Elastic, dynamically provisioned
Scalability	Manual, hardware upgrade	Automatic, in minutes
Cost	High upfront (CapEx)	Pay-as-you-go (OpEx)
Availability	Single point of failure	Multi-region redundancy
Maintenance	In-house IT team	Provider handles it

5. Explain the Cloud Computing Stack (IaaS, PaaS, SaaS) in detail. Compare all three service models with examples and suitable use cases.



IaaS – Infrastructure as a Service: Provides virtualized computing infrastructure. You get VMs, storage, and networking. You manage everything above the hypervisor.

- Examples: AWS EC2, Azure Virtual Machines, Google Compute Engine.
- Use Case: Hosting a website, running custom databases, big data processing.

PaaS – Platform as a Service: Provides a complete development + deployment environment. Provider manages infrastructure and middleware; you manage only the app.

- Examples: Google App Engine, Heroku, AWS Elastic Beanstalk.
- Use Case: Developing web/mobile apps without worrying about server setup.

SaaS – Software as a Service: Fully managed software delivered over internet. No installation or maintenance needed.

- Examples: Gmail, Salesforce, Zoom, Microsoft 365.
- Use Case: Business email, CRM, video conferencing.

Feature	IaaS	PaaS	SaaS
User manages	OS, runtime, app, data	App and data only	Nothing
Provider manages	Hardware, hypervisor	Hardware + middleware	Everything
Flexibility	Highest	Medium	Lowest
Technical skill	High	Medium	Low
Cost control	Granular	Medium	Subscription-based

6. Explain how Cloud Computing works, including the role of networks, protocols, and web services. Also discuss different deployment models (Public, Private, Hybrid) with advantages and use cases.

How Cloud Works:

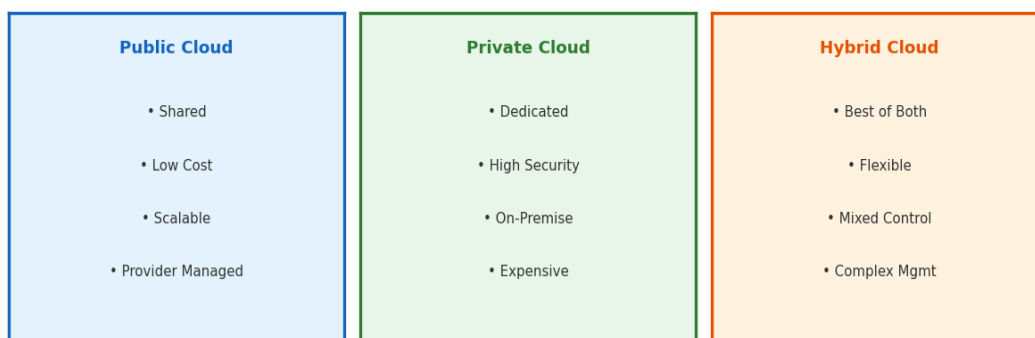
- User accesses cloud via browser/app over the internet.
- DNS resolves the cloud service address; request routed to data center.
- Load balancer distributes traffic; hypervisor allocates VM resources.
- Data served from distributed storage; response returned to user.

Role of Networks: Networks connect users to cloud resources globally. CDNs (Content Delivery Networks) cache data near users for speed. VPNs provide secure connections. Load balancers ensure high availability.

Protocols: HTTP/HTTPS for web, REST/SOAP for APIs, TCP/IP for data transport, TLS for encryption.

Web Services: Cloud exposes RESTful APIs allowing applications to programmatically use cloud features — create VMs, upload files, query databases.

Cloud Deployment Models



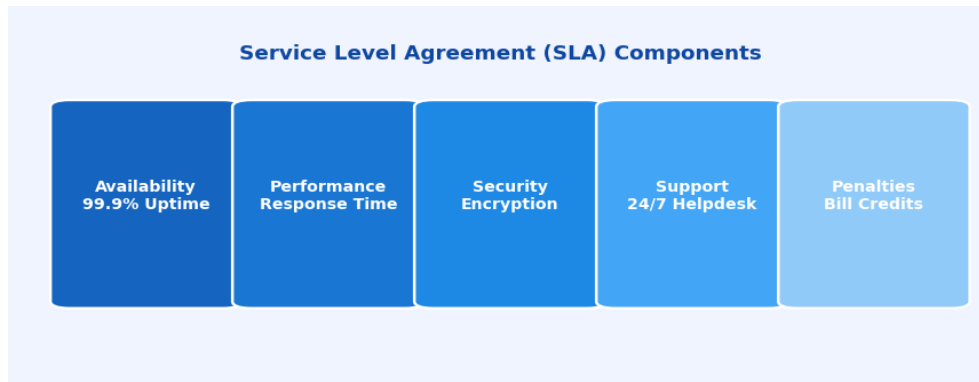
Model	Description	Advantages	Use Case
Public	Shared cloud by provider	Low cost, highly scalable	Startups, dev/test
Private	Dedicated cloud for one org	High security, full control	Banks, hospitals
Hybrid	Mix of public + private	Flexible, best of both	Regulated industries
Community	Shared among similar orgs	Cost sharing, compliance	Government agencies

7. Define Service Level Agreement (SLA) with proper diagram.

An **SLA (Service Level Agreement)** is a formal contract between a cloud provider and customer that defines the expected level of service.

Key Components of an SLA:

- **Availability:** Uptime guarantee (e.g., 99.9% = 8.7 hrs downtime/year).
- **Performance:** Response time, throughput, latency guarantees.
- **Security:** Data protection, encryption, compliance standards.
- **Support:** Response time for incidents (e.g., 1-hour critical issue response).
- **Penalties:** Credits given if SLA is breached (e.g., 10% bill credit for every 1% downtime above threshold).
- **Termination:** Conditions under which the agreement can end.

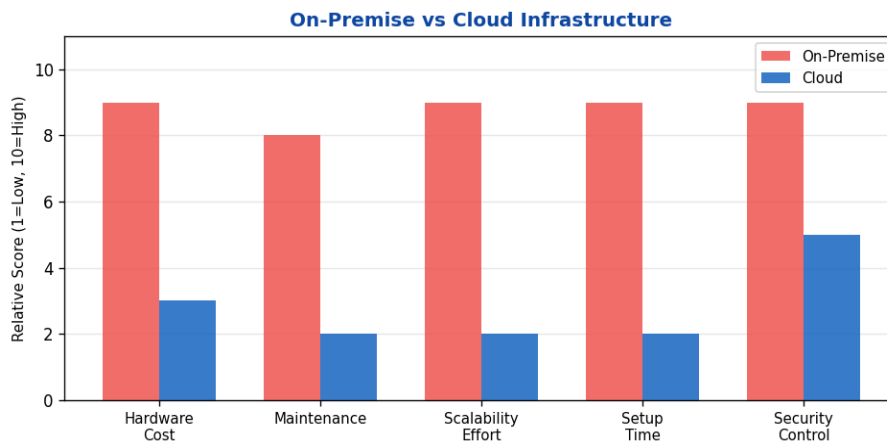


8. Discuss large-scale data processing in the cloud with suitable examples.

Cloud enables processing massive datasets using distributed computing frameworks:

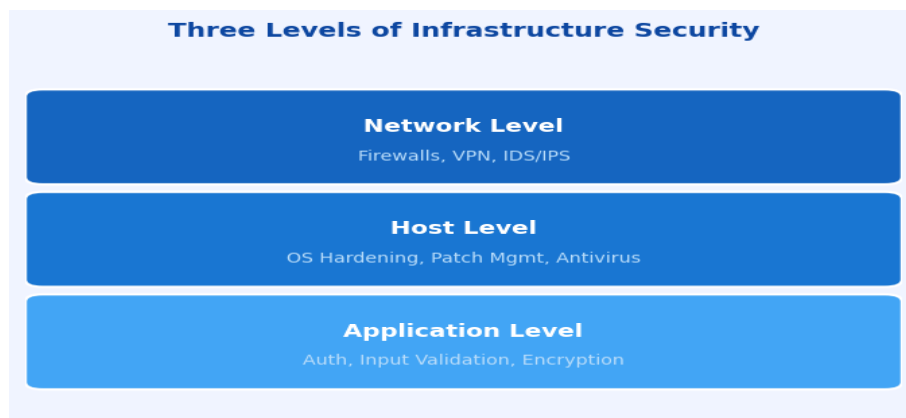
- **MapReduce:** Splits large data into chunks (Map), processes each in parallel, then combines results (Reduce). Example: Google processes web index with MapReduce.
- **Apache Hadoop:** Open-source framework for distributed storage (HDFS) and processing (MapReduce). Example: Facebook stores petabytes of user data on Hadoop.
- **Apache Spark:** Faster in-memory processing. Example: Netflix uses Spark for real-time recommendations.
- **Cloud Data Warehouses:** Amazon Redshift, Google BigQuery — query petabytes of data in seconds using SQL.
- **Streaming:** AWS Kinesis, Apache Kafka — process data as it arrives in real-time. Example: Stock market feeds, IoT sensor data.

9. Compare on-premise infrastructure vs cloud infrastructure in terms of cost, scalability, and maintenance.



Aspect	On-Premise	Cloud Infrastructure
Capital Cost	Very high (buy servers, network gear)	Zero (no hardware purchase)
Operational Cost	Power, cooling, staff salaries	Pay only for usage
Scalability	Slow — order, ship, install hardware	Instant — click and scale
Maintenance	In-house IT team required	Provider handles hardware/OS
Security Control	Full control	Shared responsibility model
Disaster Recovery	Expensive DR site needed	Built-in multi-region backups
Time to Deploy	Weeks to months	Minutes to hours
Best For	Regulated industries with static needs	Dynamic, growing businesses

10. Discuss in detail the three levels of infrastructure security: Network-level, Host-level, Application-level.



1. Network-Level Security: Protects the communication layer.

- **Firewalls:** Block unauthorized traffic based on IP/port rules.
- **VPN (Virtual Private Network):** Encrypts data in transit between user and cloud.
- **IDS/IPS:** Intrusion Detection/Prevention Systems monitor for attacks.
- **DDoS Protection:** Absorbs and mitigates distributed denial-of-service attacks.
- **Network Segmentation:** Isolates sensitive resources in separate subnets.

2. Host-Level Security: Protects individual servers/VMs.

- **OS Hardening:** Disable unused services, apply security benchmarks.
- **Patch Management:** Regular updates to fix security vulnerabilities.
- **Antivirus/Antimalware:** Detect and remove malicious software.
- **Log Monitoring:** Monitor system logs for suspicious activity.
- **Least Privilege:** Each process runs with minimum needed permissions.

3. Application-Level Security: Protects the software/application layer.

- **Authentication & Authorization:** Multi-factor auth, RBAC (Role-Based Access Control).
- **Input Validation:** Prevent SQL injection, XSS, CSRF attacks.
- **Encryption:** TLS/HTTPS for data in transit; AES encryption for data at rest.
- **Session Management:** Secure session tokens, timeout policies.

- **API Security:** OAuth 2.0, API keys, rate limiting.

Level	Focus	Key Tools/Techniques
Network	Traffic & communication	Firewall, VPN, IDS/IPS, DDoS protection
Host	Server/VM security	OS hardening, patching, antivirus, logs
Application	Software security	Auth, encryption, input validation, API security